

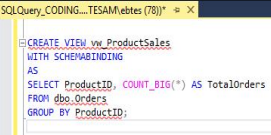
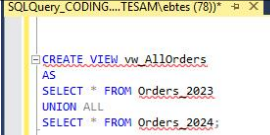


SQL Views – Research & Implementation Task

Ebtesam obaid nasser alzaabi

Part 1: Research & Documentation

Types of Views in SQL Server:

Type	Standard View	Indexed View	Union View
<u>What is it?</u>	is a virtual table created using a SELECT statement. It does not store data physically .	that stores data physically after a unique clustered index is created on it.	A Partitioned View combines data from multiple tables using <code>UNION ALL</code> . Each table holds a portion of the data , often split by date or range.
<u>Key differences</u>	• No physical storage (unlike Indexed View). • No indexing on the view itself. • Simple to create and maintain. • Query is executed every time the view is used.	Data is stored on disk Requires WITH SCHEMABINDING Faster query performance for aggregations Slower DML operations on base tables	Data remains in multiple tables Used for horizontal partitioning Improves manageability rather than raw performance Requires identical table structures
<u>Real-life use cases</u>	 <pre>CREATE VIEW vw_CustomerAccounts AS SELECT * FROM account_number, customer_id, balance, date_creation FROM Account_table;</pre>	 <pre>CREATE VIEW vw_ProductSales WITH SCHEMABINDING AS SELECT ProductID, COUNT_BIG(*) AS TotalOrders FROM dbo.Orders GROUP BY ProductID;</pre>	 <pre>CREATE VIEW vw_AllOrders AS SELECT * FROM Orders_2023 UNION ALL SELECT * FROM Orders_2024;</pre>
<u>Limitations and performance</u>	* Slower for complex queries on large tables * Performance depends entirely on base table indexes * Cannot significantly improve query speed * Complex standard views may not be datable	Extra storage required INSERT/UPDATE/DELETE operations become slower Many restrictions (no SELECT * JOIN functions) Best suited for read-heavy systems	• All tables must have the same schema • Queries can be slower if partitions are not properly filtered • Requires CHECK constraints for partition elimination • Maintenance complexity increases with many tables

1. Can We Use DML (INSERT, UPDATE, DELETE) on Views?

Yes, but not all views allow DML operations.

Whether DML is allowed depends on the type of view and how it is defined.

Which Types of Views Allow DML Operations?

Standard Views

DML Allowed: INSERT, UPDATE and DELETE

View must reference only one base table and contain no complex clauses.

Indexed Views

DML Allowed: Direct DML on the view is NOT allowed

You modify the base tables, not the indexed view.

SQL Server automatically updates the indexed view internally.

Partitioned Views

DML Allowed: Limited

INSERT is allowed only if SQL Server can determine the correct base table.

Requires CHECK constraints on base tables.

UPDATE or DELETE may fail if row mapping is unclear.

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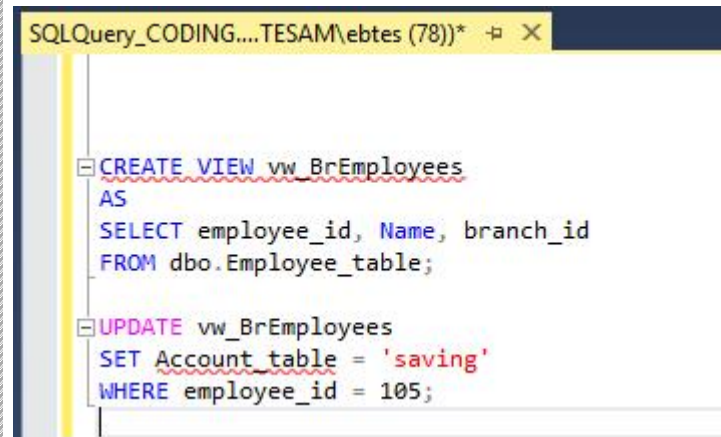
Restrictions & Limitations When Performing DML on Views

A view is NOT updatable if it contains: GROUP By - Aggregate functions (SUM, COUNT, etc.)

DML Rules for Views you should:

- Must reference only one base table
- All NOT NULL columns without defaults must be included
- Columns being modified must come directly from the base table
- No ambiguity in row identification

one real-life example



```
SQLQuery_CODING....TESAM\ebtes (78))*  X
```

```
CREATE VIEW vw_BrEmployees
AS
SELECT employee_id, Name, branch_id
FROM dbo.Employee_table;

UPDATE vw_BrEmployees
SET Account_table = 'saving'
WHERE employee_id = 105;
```

2. How Can Views Simplify Complex Queries?

Data is normalized across many tables Queries often require multiple JOIN
The same long query gets repeated by:

- Reports
- Applications
- Support teams

Part 2: Real-Life Implementation Task (Banking System):

```
SQLQuery1.sql - EB...BTESAM\ebtes (58))* X SQLQuery_CODING...TESAM\ebtes (78))*  
CREATE DATABASE bank_vi;  
use bank_vi;  
  
-----TABLE Customer -----  
  
CREATE TABLE Customer (  
    CustomerID INT PRIMARY KEY,  
    FullName NVARCHAR(100),  
    Email NVARCHAR(100),  
    Phone NVARCHAR(15),  
    SSN CHAR(9)  
);  
  
INSERT INTO Customer(CustomerID, FullName, Email, Phone, SSN)  
VALUES  
    (101, 'Ebtesam_Alzaabi', 'Ebtalzaabi@gmail.com', '977001122', '10'),  
    (104, 'Amani_Alsalhi', 'AmAlsalhi@gmail.com', '977778899', '20'),  
    (102, 'Fatma_almamri', 'Fatma@gmail.com', '977334455', '30'),  
    (103, 'Omar_alhosani', 'Omalhosani@gmail.com', '977556677', '40'),  
    (105, 'Maryam_Alzaabi', 'MarAlzaabi@gmail.com', '977778899', '50');  
  
SELECT * FROM Customer;
```

100 %

Results Messages

	CustomerID	FullName	Email	Phone	SSN
1	101	Ebtesam_Alzaabi	Ebtalzaabi@gmail.com	977001122	10
2	102	Fatma_almamri	Fatma@gmail.com	977334455	30
3	103	Omar_alhosani	Omalhosani@gmail.com	977556677	40
4	104	Amani_Alsalhi	AmAlsalhi@gmail.com	977778899	20
5	105	Maryam_Alzaabi	MarAlzaabi@gmail.com	977778899	50

SQLQuery1.sql - EB...BTESAM\ebtes (58))* -> X SQLQuery_CODING....TESAM\ebtes (78))*

-----TABLE Account-----

```
CREATE TABLE Account (  
    AccountID INT PRIMARY KEY,  
    CustomerID INT FOREIGN KEY REFERENCES Customer(CustomerID),  
    Balance DECIMAL(10, 2),  
    AccountType VARCHAR(50),  
    AccountStatus VARCHAR(20)  
);  
  
INSERT INTO Account (AccountID, CustomerID, Balance, AccountType, AccountStatus)  
VALUES  
    (100, 101, 5000.00, 'savings', 'paid'),  
    (200, 102, 1200.00, 'checking', 'paid'),  
    (300, 103, 8000.00, 'savings', 'unpaid'),  
    (400, 104, 300.00, 'checking', 'unpaid'),  
    (500, 105, 15000.00, 'savings', 'unpaid');  
  
SELECT * FROM Account;
```

100 %

Results Messages

	AccountID	CustomerID	Balance	AccountType	AccountStatus
1	100	101	5000.00	savings	paid
2	200	102	1200.00	checking	paid
3	300	103	8000.00	savings	unpaid
4	400	104	300.00	checking	unpaid
5	500	105	15000.00	savings	unpaid

SQLQuery1.sql - EB...BTESAM\ebtes (58))* -> X SQLQuery_CODING....TESAM\ebtes (78))*

-----TABLE Transaction-----

```
CREATE TABLE Transaction_table (  
    TransactionID INT PRIMARY KEY,  
    AccountID INT FOREIGN KEY REFERENCES Account(AccountID),  
    Amount DECIMAL(10, 2),  
    transaction_type VARCHAR(10), -- Deposit, Withdraw  
    TransactionDate DATETIME  
);  
  
INSERT INTO Transaction_table (TransactionID, AccountID, Amount, transaction_type, TransactionDate)  
VALUES  
    (901, 100, 500.000, 'Deposit', '2025-01-05'),  
    (902, 200, 200.000, 'Withdrawal', '2025-01-10'),  
    (903, 300, 1000.000, 'Deposit', '2025-05-12'),  
    (904, 400, 150.00, 'Withdrawal', '2025-07-20'),  
    (905, 500, 3000.00, 'Deposit', '2025-08-01');  
  
SELECT * FROM Transaction_table;
```

100 %

Results Messages

	TransactionID	AccountID	Amount	transaction_type	TransactionDate
1	901	100	500.00	Deposit	2025-01-05 00:00:00.000
2	902	200	200.00	Withdrawal	2025-01-10 00:00:00.000
3	903	300	1000.00	Deposit	2025-05-12 00:00:00.000
4	904	400	150.00	Withdrawal	2025-07-20 00:00:00.000
5	905	500	3000.00	Deposit	2025-08-01 00:00:00.000

-----TABLE Loan-----

```
CREATE TABLE Loan (  
  LoanID INT PRIMARY KEY,  
  CustomerID INT FOREIGN KEY REFERENCES Customer(CustomerID),  
  LoanAmount DECIMAL(12, 2),  
  LoanType VARCHAR(50),  
  loanStatus VARCHAR(20)  
);  
  
INSERT INTO Loan (LoanID, CustomerID, LoanAmount, LoanType, loanStatus)  
VALUES  
  (111, 101, 25000.00, 'Home Loan', 'approved'),  
  (222, 102, 10000.00, 'Car Loan', 'closed'),  
  (333, 103, 5000.00, 'Personal Loan', 'past due'),  
  (444, 104, 4000.00, 'Car Loan', 'approved'),  
  (555, 105, 2000.00, 'Personal Loan', 'closed');  
  
SELECT * FROM Loan;
```

100 %

Results Messages

	LoanID	CustomerID	LoanAmount	LoanType	loanStatus
1	111	101	25000.00	Home Loan	approved
2	222	102	10000.00	Car Loan	closed
3	333	103	5000.00	Personal Loan	past due
4	444	104	4000.00	Car Loan	approved
5	555	105	2000.00	Personal Loan	closed

Part 3: View Creation Scenarios

1- Customer Service View

```
SQLQuery1.sql - EB...BTESAM\ebtes (58))* X SQLQuery_CODING....TESAM\ebtes (78))*  
Email NVARCHAR(100),  
Phone NVARCHAR(15),  
SSN CHAR(9)  
);  
  
INSERT INTO Customer(CustomerID, FullName, Email, Phone, SSN)  
VALUES  
(101, 'Ebtesam_Alzaabi', 'Ebtalzaabi@gmail.com', '977001122', '10'),  
(104, 'Amani_Alsalhi', 'AmAlsalhi@gmail.com', '977778899', '20'),  
(102, 'Fatma_almamri', 'Fatma@gmail.com', '977334455', '30'),  
(103, 'Omar_alhosani', 'Omalhosani@gmail.com', '977556677', '40'),  
(105, 'Maryam_Alzaabi', 'MarAlzaabi@gmail.com', '977778899', '50');  
  
SELECT * FROM Customer;  
  
CREATE VIEW vw_CustomerService  
AS  
SELECT c.FullName,  
       c.Phone,  
       a.AccountStatus  
FROM Customer c  
JOIN Account a  
  ON c.CustomerID = a.CustomerID;  
  
SELECT * FROM vw_CustomerService;
```

100 %

Results Messages

	FullName	Phone	AccountStatus
1	Ebtesam_Alzaabi	977001122	paid
2	Fatma_almamri	977334455	paid
3	Omar_alhosani	977556677	unpaid
4	Amani_Alsalhi	977778899	unpaid
5	Maryam_Alzaabi	977778899	unpaid

2- Finance Department View

SQLQuery1.sql - EB...BTESAM\ebtes (58))* X SQLQuery_CODING....TESAM\ebtes (78))*

-----2---

```
CREATE VIEW vw_FinanceAccounts
AS
SELECT AccountID,
       Balance,
       AccountType
FROM Account;

SELECT * FROM vw_LoanOfficer;
```

100 %

Results

Messages

	LoanID	CustomerID	LoanAmount	LoanType	loanStatus
1	111	101	25000.00	Home Loan	approved
2	222	102	10000.00	Car Loan	closed
3	333	103	5000.00	Personal Loan	past due
4	444	104	4000.00	Car Loan	approved
5	555	105	2000.00	Personal Loan	closed

3- Loan Officer View

SQLQuery1.sql - EB...BTESAM\ebtes (58))* X SQLQuery_CODING....TESAM\ebtes (78))*

-----3---

```
CREATE VIEW vw_LoanOfficer
AS
SELECT LoanID,
       CustomerID,
       LoanAmount,
       LoanType,
       loanStatus
FROM Loan;

SELECT * FROM vw_LoanOfficer;
```

100 %

Results

Messages

	LoanID	CustomerID	LoanAmount	LoanType	loanStatus
1	111	101	25000.00	Home Loan	approved
2	222	102	10000.00	Car Loan	closed
3	333	103	5000.00	Personal Loan	past due
4	444	104	4000.00	Car Loan	approved
5	555	105	2000.00	Personal Loan	closed

4- Transaction Summary View

SQLQuery1.sql - EB...BTESAM\ebtes (58))* SQLQuery_CODING...TESAM\ebtes (78))*

-----4-----

```
CREATE VIEW vw_RecentTransactions
AS
SELECT AccountID,
       Amount,
       TransactionDate
FROM Transaction_table
WHERE TransactionDate >= DATEADD(DAY, -30, GETDATE());

SELECT * FROM vw_RecentTransactions;
```

100 %

Results Messages

AccountID	Amount	TransactionDate
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